

Marangoni spreading in new media art

Based on Chan and Fried (2024)

College of Chemical Engineering: University of Illinois in Chicago

CHE 457: Interfacial and Collision Phenomena

Javier Silva

Dr. Vivek Sharma

May 1, 2026

Introduction

- Artist Akiko Nakayama uses interfacial forces to create live art.
- Creating dendritic art by mixing ink and alcohol.
- Interfacial forces behind the art:
 - Surface Tension
 - Marangoni Forces
 - Viscosity

Figure 1

Akiko Artwirk (2019)



Figure 1

Akiko Nakayama live art performances

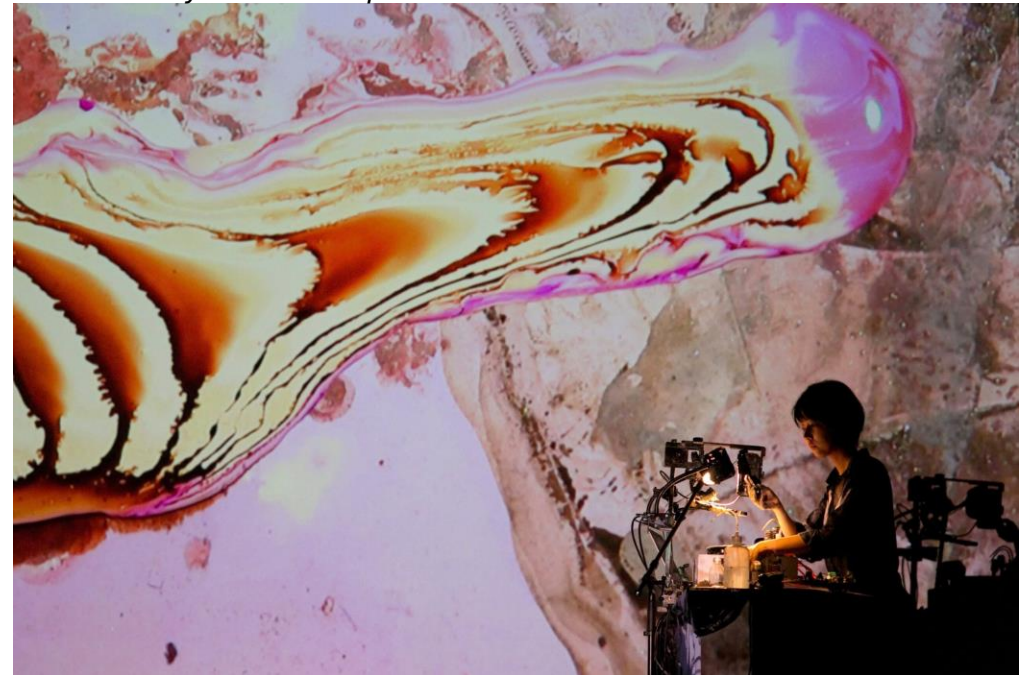


Figure 1, 2: Beppu Galleria Midobaru. (2021, August 21). Akiko Nakayama. <https://beppu-galleria-midobaru.jp/en/art/akiko-nakayama/>

Experimental System

- Mixture of ink and alcohol droplets
- Exploring Newtonian and Shear-thinning fluid behavior on substrate
- Through measuring radius over time and dendric formation behavior

Figure 4
Akiko artwork (2021)



Figure 3
Depiction of the experiment

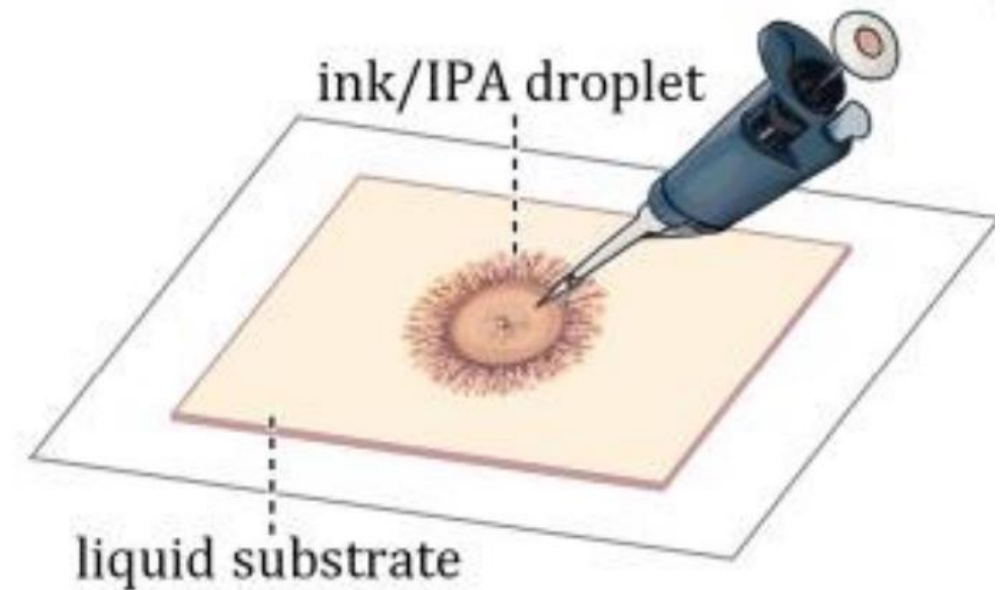


Figure 3: Chan, S. T., & Fried, E. (2024). *Marangoni spreading on liquid substrates in new media art*. **PNAS Nexus**, 3(2), 1–12.

<https://doi.org/10.1093/pnasnexus/pgae059>

Figure 4: Beppu Galleria Midobaru. (2021, August 21). *Akiko Nakayama*.

<https://beppu-galleria-midobaru.jp/en/art/akiko-nakayama>

Interfacial forces behind the art

Marangoni Forces:

- Driving force for surface tension gradient .
- Spreading is fast in the outer film and slow in the bulk center.

Viscosity:

- Driving force resisting surface tension gradient.
- Formation of the Dendric patterns.

Marangoni Effect Surface Tension
Gradient (eq.1)

$$\tau_{sur} = \nabla\gamma = \frac{\partial\gamma}{\partial C} \nabla C + \frac{\partial\gamma}{\partial T} \nabla T \text{ (eq. 1)}$$

Viscosity resistance force (eq.2)

$$\tau_{visc} = \eta \frac{\partial u}{\partial n} \text{ (eq. 2)}$$

Surface Tension Gradient and Viscosity
resistance force relation (eq.3)

$$\nabla_{sur}\gamma = \eta \left(\frac{\partial u}{\partial n} \right) \text{ (eq. 3)}$$

Results from Scaling Laws

Two distinct regions formed from the experiment regardless of liquid substrate:

- Inner Radius = $R_{in} = t^{\frac{1}{4}}$
 - Slow expansion of inner radius
- Outer Radius = $R_{out} = t^{\frac{3}{8}}$
 - Fast expansion of outer radius

Figure 7
Expansion analysis

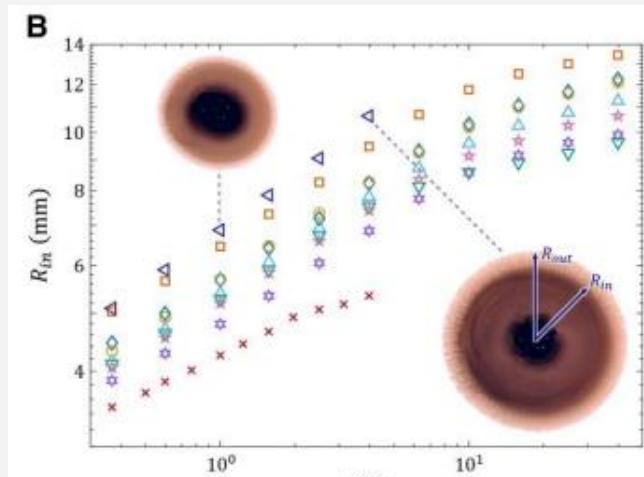


Figure 5
Inner radius expansion analysis

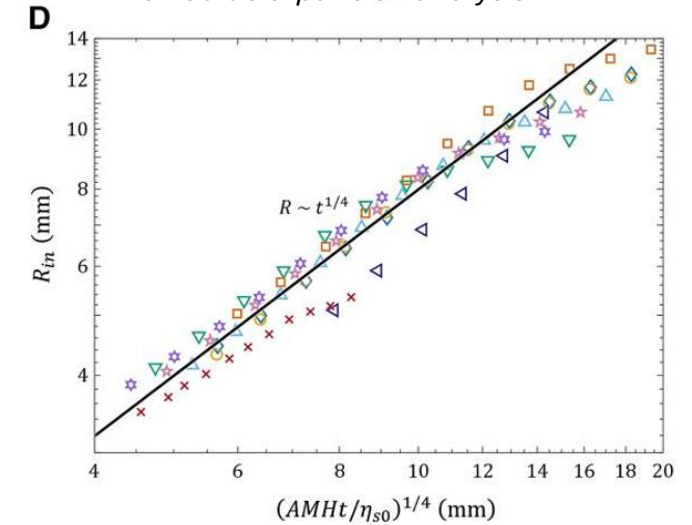


Figure 6
Outer radius expansion analysis

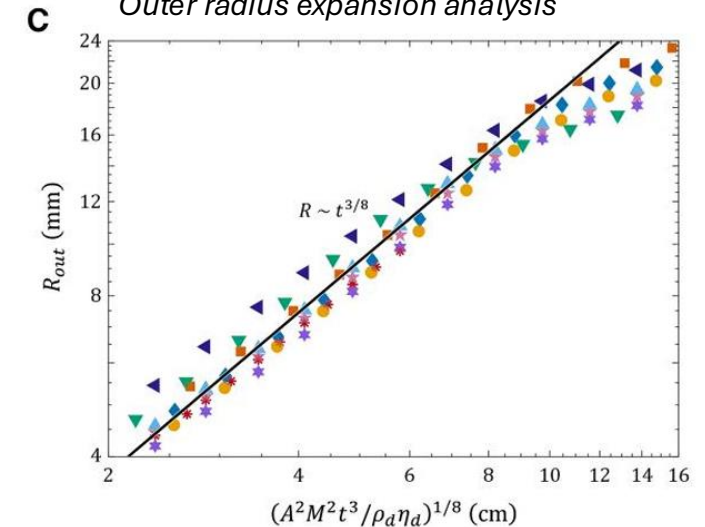


Figure 5, 6, 7: Chan, S. T., & Fried, E. (2024). Marangoni spreading on liquid substrates in new media art. **PNAS Nexus**, 3(2), 1–12.
<https://doi.org/10.1093/pnasnexus/pgae059>

Discussion (Part 1)

- Newtonian = smooth spread
- Shear-thinning = Dendritic spread behavior
- Scaling law follows the Marangoni Effect:
 - Outer (Thin film)
 - Inner (Bulk)

Figure 10

Akiko artwork (2023)



Figure 8

Marangoni effect on Newtonian liquid substrate

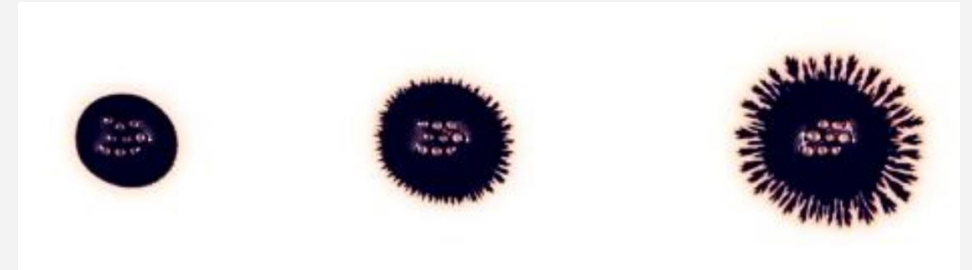


Figure 9

Marangoni effect on Shear-thinning liquid substrate

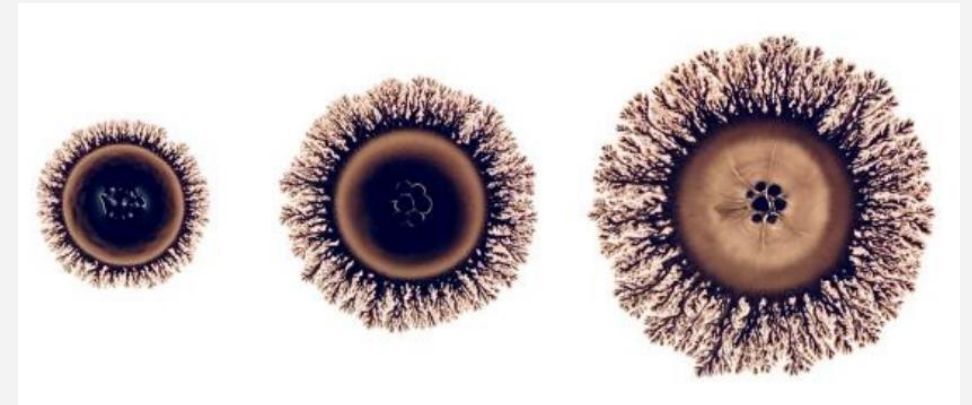


Figure 8, 9: Chan, S. T., & Fried, E. (2024). *Marangoni spreading on liquid substrates in new media art*. *PNAS Nexus*, 3(2), 1–12.
<https://doi.org/10.1093/pnasnexus/pgae059>

Figure 10: Beppu Galleria Midobaru. (2021, August 21). Akiko Nakayama.
<https://beppu-galleria-midobaru.jp/en/art/akiko-nakayama>

Discussion (Part 2)

Critique

- Assumption were oversimplified for perfect conditions

Further Analysis

- Explore Marangoni effect in the coating and paint industry applications

Figure 12
Akiko artwork (2021)



Figure 11

Marangoni effect considered in the development of coats used in industry



Figure 11: Thermal Spray. (n.d.). *All you need to know about metal film coatings.* <https://www.thermalspray.com/all-you-need-to-know-about-metal-film-coatings/>

Figure 12: Beppu Galleria Midobaru. (2021, August 21). Akiko Nakayama. <https://beppu-galleria-midobaru.jp/en/art/akiko-nakayama>

Conclusion

- Marangoni forces can be used in art and in engineering at the same time
- Marangoni forces and viscosity determine the shape and pattern of the art

Figure 14
Akiko Artwork (2024)



Figure 13
Akiko Nakayama live art performance



Figure 13,14: Beppu Galleria Midobaru. (2021, August 21). Akiko Nakayama. <https://beppu-galleria-midobaru.jp/en/art/akiko-nakayama/>

References

- Beppu Galleria Midobaru. (2021, August 21). *Akiko Nakayama*.
<https://beppu-galleria-midobaru.jp/en/art/akiko-nakayama/>
- Chan, S. T., & Fried, E. (2024). *Marangoni spreading on liquid substrates in new media art*. **PNAS Nexus**, 3(2), 1–12.
<https://doi.org/10.1093/pnasnexus/pgae059>
- Thermal Spray. (n.d.). *All you need to know about metal film coatings*.
<https://www.thermalspray.com/all-you-need-to-know-about-metal-film-coatings/>